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Attention deficit/hyperactivity disorder diagnosis in adulthood: Prevalence and predictive utility of neuropsychological scales

Maria Pleshkevich[1,2], Robiann Broomfield[1], Sydney Burton[1,3,†], Gabriela Castro[1,3,†], Alyssa G. Gill[1,3,†], Matthew Jerram[2], Setari Parsa[1,*]

[1]Department of Psychiatry, Harvard Medical School, Beth Israel Deaconess Medical Center, Boston, MA 02215, USA
[2]Department of Psychology, Suffolk University, Boston, MA 02108, USA
[3]Department of Clinical Psychology, William James College, Newton, MA, 02459, USA
*Corresponding author at: Department of Psychiatry, Neuropsychological Assessment Center, Beth Israel Deaconess Medical Center, 185 Pilgrim Rd., Boston, MA 02215, USA. Tel.: (617) 632-0908. *Email address*: sparsa@bidmc.harvard.edu (S. Parsa).

†Sydney Burton, Gabriela Castro and Alyssa G. Gill contributed equally to this work.

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ABSTRACT

Objective: This cross-sectional study investigated the prevalence of adult Attention Deficit/Hyperactivity Disorder (ADHD) evaluations in an outpatient neuropsychology clinic and the sensitivity and specificity of neuropsychological assessments in differentiating ADHD with comorbid psychiatric conditions from psychiatric-related cognitive deficits.

Method: Logistic regression analyses were run on neuropsychological testing results extracted from medical records to evaluate each test's ability to predict ADHD diagnosis. Receiver Operating Characteristic curves were plotted, and an Area Under the Curve (AUC) >0.7 was considered acceptable.

Results: Sixty-seven of 252 reviewed charts were included ($n = 30$ with ADHD, $n = 37$ without ADHD). Overall, 54% of individuals presenting to the clinic were referred for ADHD, and 58% of those diagnosed with ADHD had a comorbid psychiatric disorder. No cognitive testing was predictive of ADHD diagnosis (AUC range: 0.503–0.636). The Generalized Anxiety Disorder-7 was the strongest predictive scale of ADHD (AUC = 0.800; $\chi^2(28) = 8.216$, $p = .004$, Nagelkerke $R^2 = 0.333$). Adults who endorsed higher levels of anxiety were 28% more likely to be diagnosed with ADHD (OR = 1.28, 95%CI [0.05, 0.45]). Subtests of ADHD scales performed worse, but still within the adequate range (AUC range: 0.703–0.751). Sample demographics were notable for over-representation of women ($n = 46$, 69%), and they skewed White ($n = 43$, 64%) and highly educated (M years of education = 15.79, $SD = 1.94$), which may limit generalizability.

Conclusions: Cognitive testing profiles and screeners of current inattention and hyperactivity were indiscriminate between individuals with ADHD and comorbid psychiatric conditions versus psychiatric conditions alone. Current anxiety was the strongest predictor of adult ADHD diagnosis.

Keywords: Attention Deficit/Hyperactivity Disorder (ADHD), anxiety, depression, executive functions, assessment

Introduction

Attention Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder marked by persistent and functionally impairing symptoms of inattention and/or hyperactivity and impulsivity (American Psychiatric Association [APA], 2022). Inattentive symptoms may include forgetfulness in daily activities, such as while running errands or in keeping appointments, avoidance of tasks requiring sustained attention, and difficulty with organization (American Psychiatric Association [APA], 2022). Hyperactive symptoms may include fidgeting, difficulty engaging in quiet leisure, and talking excessively (American Psychiatric Association [APA], 2022). Inattention, hyperactivity, and impulsivity are considered deficits in executive functions, as they are directly related

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to cognitive processes such as working memory, inhibitory control, and cognitive flexibility, which regulate attention, behavior, and impulse control (Barkley, 1997; Nigg, 2005).

Although ADHD symptoms must be present prior to age 12 to meet Diagnostic Statistical Manual-5-Text Revised (DSM-5-TR) criteria for diagnosis, many individuals with ADHD may not be diagnosed until adulthood (Rivas-Vazquez et al., 2023). While estimates of ADHD prevalence rates in adulthood range from 2.5% to 4.4% (American Psychiatric Association [APA], 2022), a meta-analysis found that among epidemiological research on ADHD, very little research exists using samples of adults >25 years old (Willcutt, 2012). Thus, existing studies on ADHD prevalence may not generalize across the lifespan. Moreover, despite worldwide prevalence rates falling below 5%, a recent meta-analysis found that as many as 26.7% of adults presenting to outpatient psychiatric clinics were diagnosed with ADHD (Adamis et al., 2022). This finding highlights the high prevalence of ADHD presentations within psychiatric settings and underscores the importance of accurate assessment and differential diagnosis in adulthood.

ADHD in adulthood is highly comorbid with psychiatric disorders. A World Health Organization survey of 15,991 first year college students found an ADHD prevalence of 15.9%, of which 58.4% had psychiatric comorbidities (Mak et al., 2022). Internalizing disorders were most common (36.0%), followed by bipolar disorder (11.3%), and externalizing disorders (9.8%; Mak et al., 2022). Gerhand and Saville's (2022) systematic review of research published in adult samples concluded that ADHD may be underdiagnosed in a psychiatric population. Their review found prevalence rates ranging from 6.9 to 38.75%, significantly higher than the estimated 2.8% ADHD prevalence in the general adult population.

ADHD diagnosis in adulthood poses certain challenges to appropriately identifying the disorder. Retrospective studies of childhood ADHD symptom recall have demonstrated recall bias (Katzman et al., 2017; Taylor et al., 2022; Von Wirth et al., 2021). Vitola et al. (2017) found that DSM-5-TR criteria were more likely to yield an ADHD diagnosis among adult samples compared to DSM-IV criteria, attributing this effect to the higher relevance of inattentive phenotypes in adult populations. It is important to note, however, that the DSM-5-TR criteria were developed to better address the underdiagnosis of ADHD in adulthood (Rivas-Vazquez et al., 2023). Nonetheless, an important diagnostic criterion is that inattentive/hyperactive symptoms are not better explained by another mental disorder (American Psychiatric Association [APA], 2022). The high prevalence of comorbid psychiatric conditions with ADHD in adulthood can further muddle the clinical picture, as many psychiatric diagnoses, such as depression (Keller et al., 2019), bipolar disorder (Camelo et al., 2013), and anxiety disorders (Volel et al., 2018), present with impairments in executive functioning.

An additional challenge in adult ADHD assessment involves the potential influence of symptom exaggeration, secondary gain, or noncredible self-reporting, particularly in contexts where diagnosis may provide access to stimulant medication, academic accommodations, or occupational support services (Becke et al., 2021; Sadek, 2022). Prior literature has demonstrated rates as high as 36.6% of symptom validity test (SVT) and 49.3% of performance validity test (PVT) failure among adults referred for ADHD evaluations (Hirsch et al., 2022). Although meta-analyses have found low rates of PVT failure among individuals with psychiatric diagnoses such as depression and anxiety (Legemaat et al., 2023; McWhirter et al., 2020), less research exists on SVTs failure rates in psychiatric populations. However, one study investigating SVT failure rate in Veterans with combat exposure found that 78% of individuals who failed an SVT had at least one psychiatric diagnosis, compared to 46% without (Ord et al., 2021). In the context of ADHD, Woloszyn et al. (2026) found that perceived stress was associated with increases in ADHD symptom reporting, and that removing individuals with failed SVTs from their analysis resulted in depression and anxiety symptoms no longer being associated with ADHD symptoms reported in childhood. This suggests that ADHD comorbidity with psychiatric disorders could amplify risk of PVT or SVT failure. In all, these findings emphasize the importance of incorporating validity assessment procedures into diagnostic formulations, particularly among individuals with ADHD and comorbid psychiatric conditions.

At the same time, clinicians must balance consideration of symptom validity with the recognition that psychiatric distress, executive dysfunction, and developmental histories may legitimately contribute to attentional complaints. Moreover, delayed recognition of ADHD symptoms in adulthood may be associated with poorer long-term psychiatric and functional outcomes (Holden & Kobayashi-Wood, 2025; Katzman et al., 2017; Long & Coats, 2022). Persistent attentional and executive functioning difficulties may contribute to impairments across academic, occupational, interpersonal, and emotional domains, while delayed identification may limit access to appropriate interventions and supports. Importantly, qualitative analyses have demonstrated that identification of ADHD has important implications on quality of life, such as increased self-efficacy via feelings of relief at seeking appropriate and targeted treatment and no longer viewing one's life as an "unnecessary struggle" (Long & Coats, 2022).

Present study

Although adult diagnosis of ADHD has been gaining more attention in recent years (Rivas-Vazquez et al., 2023), little research is available in this population (Willcutt, 2012). Psychiatric clinics experience a high rate of adults presenting with concerns for ADHD (Adamis et al., 2022). Similar symptoms to ADHD are present across a myriad of psychiatric disorders, which can make differential diagnosis difficult. Given the difficulty of assessing ADHD in adulthood due to the high prevalence and confounding influences of psychiatric comorbidities (Waltereit et al., 2023), it is important for researchers to investigate the clinical utility of existing standard-of-care neuropsychological assessments used in the diagnosis of ADHD. While studies have investigated the utility of neuropsychological tests in differentiating individuals with ADHD from healthy and clinical controls (for a review, see Pettersson et al., 2018), to our knowledge, no study has specifically compared individuals with ADHD and comorbid

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psychiatric conditions to individuals with psychiatric conditions alone.

There are important treatment implications for correctly differentiating ADHD from cognitive sequelae secondary to psychiatric diagnoses alone. Adults diagnosed with ADHD are more likely to be treated using stimulants than individuals diagnosed in childhood (Young & Goodman, 2016). While stimulants may help treat ADHD symptoms, they may be ineffective for executive functioning deficits associated with psychiatric etiologies. For example, while they may provide initial relief in the treatment of depression, these effects have been found to be short-lived, pointing to a potentially different biomechanism underlying attention deficits in depression (Malhi et al., 2016). On the other hand, underdiagnosing ADHD may contribute to worse long-term psychiatric trajectories (Katzman et al., 2017).

The present study aims to address gaps in the knowledge of how ADHD and comorbid psychiatric conditions may affect cognition, as well as identify the utility of standard-of-care assessments in differentiating individuals diagnosed with ADHD from those diagnosed with other psychiatric disorders. With the use of available medical records, the aims of the study are as follows: (a) to present demographic and neuropsychological data of individuals presenting for an ADHD evaluation to an adult outpatient neuropsychology clinic within the psychiatry department of an urban academic medical center, (b) to describe characteristics of individuals diagnosed with comorbid ADHD and psychiatric disorders versus those not diagnosed with ADHD, and (c) to evaluate the sensitivity and specificity of standard-of-care neuropsychological assessments in differentiating ADHD comorbid with psychiatric diagnoses versus psychiatric related cognitive deficits alone.

Method

Setting

Neuropsychological assessment records were reviewed of all individuals presenting to the clinic from July 1, 2024 through May 31, 2025. The outpatient clinic was situated in an academic medical center in a populous city in the Northeastern United States. The clinic has two neuropsychologists on staff. Assessments are completed by trainees (practicum students enrolled in PhD and PsyD clinical psychology programs, American Psychological Association accredited program interns, and postdoctoral students) under the supervision of staff neuropsychologists. Each neuropsychologist sees two to five cases per week, with one patient slot per provider being reserved for ADHD evaluations. The clinic requires referrals for evaluation by a medical provider and accepts private and public medical insurance, including state and federal plans.

Evaluations involve 2–4 hours of in-person cognitive testing using a variety of modalities, such as hands-on tasks, auditory tasks, visual tasks, and pencil and paper tasks. Domains measured include executive functions (processing speed, attention span, working memory, problem-solving approaches), verbal abilities (verbal reasoning and fluency, confrontation naming), visuospatial and visuomotor abilities (visual and spatial reasoning, hands-on assembly, fine motor dexterity), and memory (word-lists, stories, and visual). Self-reported surveys on child and current behavioral functioning and current psychological functioning are additionally completed. Testing batteries are created on a case-by-case basis, depending on each patient's unique presenting concerns and differential diagnoses. The order of tests administered is additionally varied per patient. Trainees have been instructed not to administer tasks that could interfere with memory encoding and retrieval during delays between memory test trials. For example, a visuospatial task would not be administered during the delay of a visual memory task.

Participants

Individuals included in the study were those who presented to the clinic with a differential diagnosis question of attention deficit hyperactivity disorder versus psychiatric related cognitive deficits. Certain individuals were excluded to ensure that the sample reflected the study's goal of examining the utility of common neuropsychological assessments in differential diagnosis. Exclusion criteria were as follows: referral reason other than ADHD; absence of a primary or comorbid psychiatric diagnosis; Intelligence Quotient (IQ) ≤ 70 in order to minimize confounding effects of global intelligence; age restrictions (<18 or ≥ 65) were applied to maintain consistency with the adult population and to minimize effects of aging; participants with untreated sleep apnea, epilepsy/seizures, traumatic brain injury within three years, or suspected neurodegenerative conditions due to the potential for such medical conditions to cause cognitive deficits that mimic/obscure ADHD symptoms; current stimulant treatment at the time of testing in order to avoid medication-related enhancements on testing; invalid test results based on embedded and stand-alone PVTs; or absence of a data summary attached to the neuropsychological testing report. Standard practice in the clinic from which data were collected is to not report raw data from tests in which individuals failed SVTs—thus, data from these measures was not included by the nature of the scores available in chart review. However, neuropsychological data was still included, as research suggests that SVT and PVT failure are not strongly concordant (Benowitz et al., 2025; Ovsiew et al., 2023).

Individuals with ADHD and no comorbid psychiatric diagnosis were excluded from the sample, as the primary research question was to investigate whether standard of care neuropsychological tests are specifically sensitive and specific to ADHD psychiatric comorbidity. Sensitivity and specificity analyses are well-documented in other studies on adults with ADHD (Rosso et al., 2023; Wasserman & Wasserman, 2012), but less is known about how tests perform in individuals with both ADHD and a comorbid psychiatric diagnosis. Existing studies have combined individuals with ADHD with and without psychiatric comorbidity into one diagnostic group, with the resulting samples having less psychological burden than their respective clinical control groups (Guo et al., 2021; Onandia-Hinchado et al., 2021; Pettersson et al., 2018).

Data collection

The present study was conducted via retrospective chart review. Available medical records of all individuals presenting for neuropsychological assessment to an adult outpatient clinic within an urban academic medical center were

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reviewed from May 1, 2025 through June 10, 2025. Due to a shift to a new electronic medical record system, records were only available starting July 1, 2024. All study activities were approved by the local Institutional Review Board (IRB). This study met criteria for exempt research, and the IRB thus approved a waiver of informed consent. The study was conducted in compliance with the IRB.

All charts were evaluated for inclusion and exclusion criteria. The number of charts meeting exclusion criteria were recorded. Data were entered from charts meeting inclusion criteria using solely the neuropsychological testing report, which detailed a narrative clinical history, including demographics, provided a data summary of neuropsychological test scores (raw and normed), and assigned a diagnosis based on a combination of clinical interview and testing outcomes. All variables included were determined a priori to reduce potential for bias. The following demographic variables were recorded: age at testing, birth sex, gender identity, race and ethnicity, handedness, and years of education. The following covariates were selected for exploratory analyses due to their potential confounding effects on cognition and/or the testing environment: provider seen for the evaluation, presence of vascular risk factors (defined as an elevated A1C of 6.5 or higher (Sola et al., 2024), high blood pressure (Hughes et al., 2020), high cholesterol (Zhao et al., 2024), smoking (Anstey et al., 2007), or obesity (Phirom et al., 2025), self-reported poor sleep quality (defined as disrupted sleep, snoring, feeling poorly rested, and self-reported hyper or hypo somnolence; Qin et al., 2023), prior substance use disorder (SUD) and/or SUD at the time of testing, SUD duration (Ramey & Regier, 2019), marijuana use at time of testing and how long the patient abstained from marijuana prior to testing (Scott et al., 2018), history of migraines (Gu et al., 2022), whether or not English was the patient's dominant language, whether or not an interpreter was used for testing (Casas et al., 2012), history of childhood trauma or neglect (Kavanaugh et al., 2017), presence of pain during testing and level of pain reported (0–10, 10 = the worst; Zhang et al., 2021), mood rating at time of testing (1–10, 10 = the best; Rock et al., 2014)), anxiety rating at time of testing (1–10, 10 = the most; Nguyen et al., 2025), and hours of sleep achieved prior to testing (Lo et al., 2012).

Each neuropsychological evaluation report provided a descriptive estimate of each patient's premorbid cognitive abilities based on performance on word reading and verbal and/or visual reasoning, depending on the patient's reported history, together with reported educational and vocational background. Descriptive estimates followed the following percentile estimates, which were converted to z-scores based on a normal distribution: exceptionally low (<3%ile; $z = -2$), below average (3%ile–8%ile; $z = -1.5$), low average (9%ile–10%ile; $z = -1$), low average to average (21%ile–32%ile; $z = -0.5$), average (33%ile–50%ile; $z = 0$), average to high average (51%ile–75%ile, $z = 0.5$), high average (76%ile–85%ile; $z = 1$), high average to above average (85%ile–91%ile; $z = 1.25$), above average (92%ile–97%ile; $z = 1.5$), and exceptionally high (>97%ile, $z = 2$). These estimates served as a reference point for interpretation of each individual's neuropsychological testing results. Raw and normed scores of each neuropsychological test administered and reported were recorded for analysis. Each normed score was converted to a z score. The premorbid estimate scores were subtracted from normed neuropsychological testing scores to create a deviation score for each neuropsychological testing variable. Positive numbers indicate where individuals performed better on a given test than their premorbid estimate, while negative numbers indicate worse than expected performance.

The following outcome variables were reported: whether or not ADHD was diagnosed, ADHD type that was diagnosed, whether or not another neurodevelopmental disorder was diagnosed, and which psychiatric comorbidities were diagnosed. Psychiatric conditions were sorted into the following categories: mood disorders, anxiety-spectrum disorders, both mood and anxiety disorders, and others. Conditions such as major depressive disorder, unspecified depressive disorder, and bipolar disorder, were coded as mood disorders. Disorders such as generalized anxiety disorder, post-traumatic stress disorder, and obsessive compulsive disorder, were coded as anxiety-spectrum disorders. Examples of other disorders include substance use disorders and adjustment disorders.

Neuropsychological and psychiatric measures

The following neuropsychological and psychiatric measures were utilized: Advanced Clinical Solutions (ACS): Test of Premorbid Functioning (TOPF; Pearson Clinical, 2009; Wechsler, 2011), Wechsler Adult Intelligence Scale–Fourth Edition (WAIS-IV; Wechsler, 2008), Conners Continuous Performance Test–3rd Edition (CPT-3; Conners, 2014), Boston Naming Test–Second Edition (BNT; Heaton et al., 2004; Kaplan et al., 2001), Neuropsychological Assessment Battery (NAB) Naming Test (Stern & White, 2003), Controlled Oral Word Association Test (FAS/Animals; Benton et al., 2017; Heaton et al., 2004; Tombaugh, 1999), Delis-Kaplan Executive Function Scale (DKEFS)–Verbal Fluency (Delis et al., 2001), Trail Making Test (TMT) A and B (Army Individual Test Battery, 1944; Heaton et al., 2004; Tombaugh, 2004), California Verbal Learning Test–Second Edition (CVLT-II; Delis et al., 2000), Wechsler Memory Scale–Fourth Edition (WMS-IV; Wechsler, 2008), Rey-Osterrieth Complex Figure Test (RCFT; Meyers & Meyers, 1995; Osterrieth, 1944; Rey, 1941; Somerville et al., 2000), Brief Visual Memory Test–Revised (BVMT-R; Benedict, 1997), Tower of London–Drexel University, 2nd Edition (TOLDX; Culbertson & Zillmer, 2005), Barkley Adult ADHD Rating Scale–IV (BAARS-IV; Barkley, 2011) symptom counts, Conners' Adult ADHD Rating Scales–Self Report: Long Version (CAARS; Conners et al., 1999), Beck Depression Inventory–Second Edition (BDI-II; Beck, 1996), Generalized Anxiety Disorder–7 (GAD-7; Spitzer et al., 2006), and Personality Assessment Inventory (PAI; Morey, 1991). Data from individuals failing SVTs on the PAI and the CAARS were not reported in neuropsychological reports and were therefore implicitly excluded. Finally, an SVT cut off of ≥36 on the BDI-II (Kent et al., 2025) was applied to the present sample, and BDI-II data from individuals scoring above this threshold were excluded from analyses.

Statistical methods

Descriptive analyses were conducted on demographic, covariate, and neuropsychological test scores and deviation

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scores, when able to be computed. They were separated by cohort: individuals diagnosed with ADHD and a comorbid psychiatric diagnosis, and individuals not diagnosed with ADHD. Means (*M*) and standard deviations (*SD*) were reported for continuous variables, while frequencies and percentages were reported for categorical variables. The data were evaluated to ensure assumptions for respective statistical methods were met. Independent samples t-tests were conducted to evaluate for statistically significant differences between cohorts for continuous variables, and Cohen's *d* was computed to evaluate effect size. Chi-square analysis was conducted on categorical variables. Fisher's exact test was conducted on variables containing <5 items in a cell. Statistical significance was set at $\alpha = 0.05$. Effect sizes greater than $d = 0.6$ would be discussed, as well as means that reflected clinical significance (i.e., neuropsychological deviation scores (z) ≤ -1.5 standard deviations from an individual's premorbid estimate and survey scores meeting clinical thresholds).

Logistic regressions were conducted for the deviation score of each neuropsychological test to determine whether the test predicted cohort membership. A Receiver Operating Characteristic (ROC) curve was plotted, and the Area Under the Curve (AUC) was used to determine the clinical utility of tests, with measures above 0.7 being considered acceptable (Hosmer et al., 2013). Depending on total available sample size, statistically significant covariates were to be included in the logistic regression. All variables included in logistic regression were assessed for normality via computing skewness and kurtosis. Corrections for family wise error were not completed, as the primary outcomes were sensitivity and specificity analyses not reliant on p-values. Variables were considered normally distributed with a skewness between −2 and 2 and kurtosis between −7 and 7. Statistical analysis was performed in JASP 0.18.3 and figures were created in https://BioRender.com and Python 3× (Python Software Foundation, n.d.).

Results

Aim 1: ADHD evaluation clinic summary

Two hundred and fifty-two individuals presented for neuropsychological evaluation from July 1, 2024 to May 31, 2025. Fifty-four percent ($n = 137$) of individuals seen at the clinic were referred for an ADHD evaluation. Of these, 16.1% ($n = 22$) were not diagnosed with a comorbid psychiatric disorder and were thus excluded from the study. Twenty-four percent ($n = 28$) were older than 65 years old. Other rates of meeting exclusion criteria are presented in Fig. 1.

The final sample included 67 participants, of which 44.8% ($n = 30$) were diagnosed with ADHD and a comorbid psychiatric disorder, while 55.2% ($n = 37$) were diagnosed with a psychiatric disorder alone. Overall, 57.7% of individuals with ADHD met criteria for another DSM-5 psychiatric disorder. Of those diagnosed with ADHD, 33.3% ($n = 10$) were diagnosed with ADHD inattentive presentation, 3.3% ($n = 1$) were diagnosed with ADHD hyperactive presentation, 33.3% ($n = 10$) were diagnosed with ADHD combined type, and 30% ($n = 9$) were diagnosed with an unspecified ADHD presentation. Demographics for each cohort are presented in Table 1. Demographics were not statistically significantly different between groups.

Forty-nine percent of individuals ($n = 33$) were diagnosed with both mood and anxiety disorders, 28.4% ($n = 19$) were diagnosed with an anxiety disorder only, and 19.4% ($n = 13$) were diagnosed with a mood disorder only. Two participants were classified under other disorders (alcohol use disorder and adjustment disorder). Among those with ADHD, 20% ($n = 6$) were diagnosed with a mood disorder, 26.67% ($n = 8$) were diagnosed with an anxiety disorder, and 50% ($n = 15$) were diagnosed with both. Anxiety disorders included generalized anxiety disorder, unspecified anxiety disorder, obsessive compulsive disorder, unspecified obsessive-compulsive and related disorder, social anxiety disorder, posttraumatic stress disorder, unspecified trauma-and-stressor-related disorder, and somatic symptom disorder. Mood disorders included persistent depressive disorder, major depressive disorder, major depressive disorder with seasonal pattern, recurrent depressive disorder, unspecified mood disorder, unspecified depressive disorder with anxiety, major depressive disorder with anxious distress, unspecified bipolar and related disorder, and bipolar II disorder. Other diagnoses included cannabis use disorder, unspecified binge eating disorder, excoriation disorder, trichotillomania, and insomnia disorder.

Aim 2: Covariates and cognitive and psychological profiles

All continuous variables included in the analyses were normally distributed, apart from one cognitive measure (CPT-3 Omissions, in which an outlier was centered to the second closest value in order to achieve normality). Individuals with and without ADHD did not differ on any of the evaluated covariates, including sleep quality (Supplementary Table 1), did not display different rates of psychiatric comorbidity (Supplementary Table 2), and had similar cognitive profiles (Supplementary Table 3). All effect sizes were < 0.6. Almost all cognitive tests were within normal limits in both groups. However, both groups on average displayed worse-than-expected performance on TMT-A (ADHD: z-score $M = 1.69$, $SD = 1.79$; non-ADHD: $M = -1.50$, $SD = 1.14$; $t(59) = 0.51$, $p = .394$, $d = 0.132$) and RCFT copy (ADHD: raw score $M = 30.07$, $SD = 4.39$; non-ADHD: $M = 31.54$, $SD = 4.23$; $t(55) = 1.28$, $p = .749$, $d = 0.207$) and immediate recall (ADHD: z-score $M = -1.69$, $SD = 1.13$; non-ADHD: $M = -1.66$, $SD = 1.30$; $t(24) = 0.05$, $p = .960$, $d = 0.020$; Fig. 2). Only individuals with ADHD performed below expectations on RCFT delayed recall (z-score $M = -2.00$, $SD = 1.03$) and CVLT-II Hits (z-score $M = -1.70$, $SD = 1.06$). However, these results were not statistically significantly discrepant from the results of those without ADHD, and the effect sizes were considered small to medium ($t(21) = 1.22$, $p = .237$, $d = 0.51$; $t(60) = 1.65$, $p = .607$, $d = 0.422$, respectively).

Individuals diagnosed with ADHD had statistically significantly higher scores of childhood inattention ($M = 5.87$, $SD = 2.38$) and hyperactivity ($M = 5.13$, $SD = 3.42$) symptom counts on the BAARS-IV than individuals without ADHD (inattention: $M = 3.77$, $SD = 2.92$, $t(52) = 2.82$, $p = .007$,

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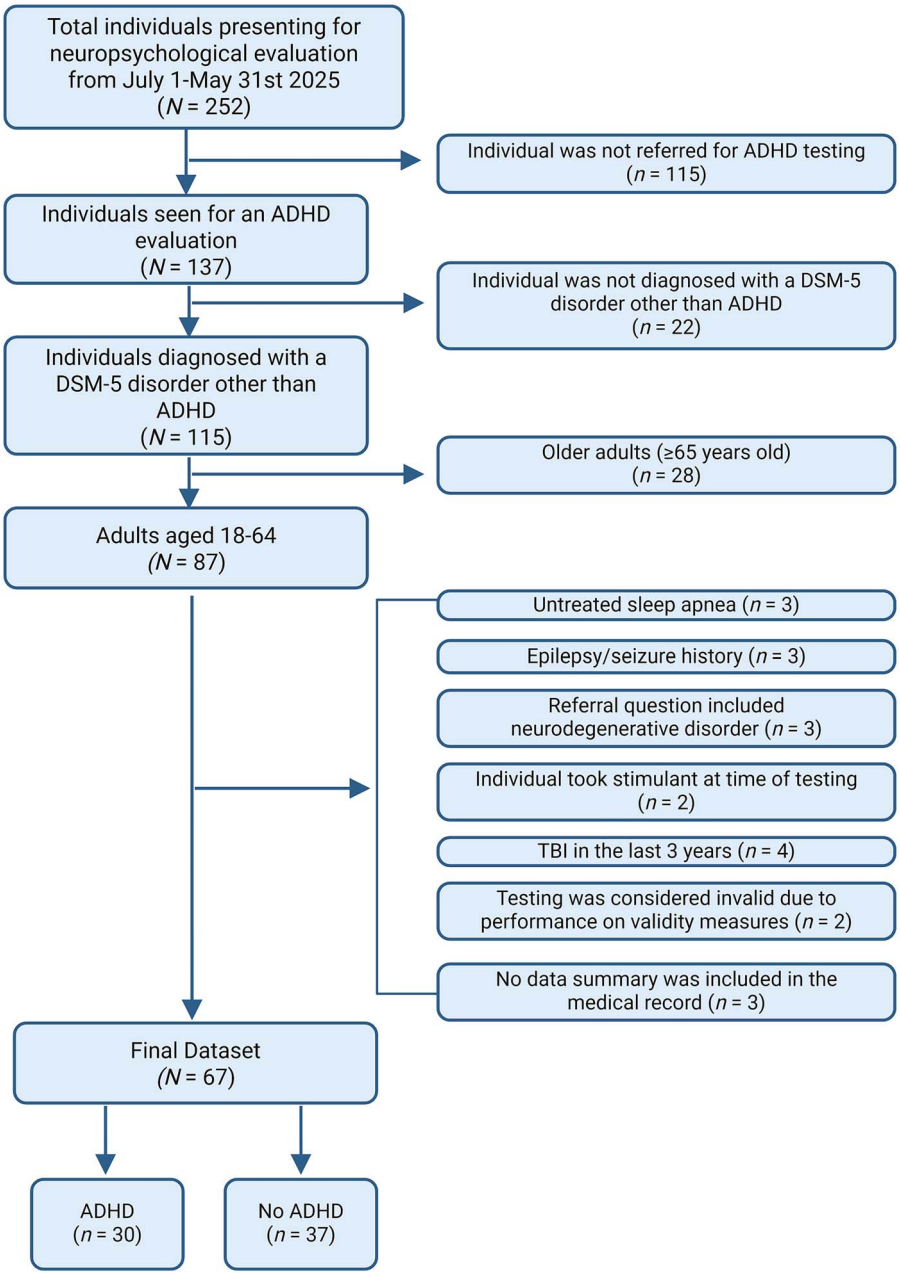


Figure 1 The present figure provides a flow chart visual of the number of participants that were determined ineligible at each stage of record review, resulting in the final sample included in the analyses of this study. i Created in BioRender. Pleshkevich, M. (2026) https://BioRender.com/871bx5f.

$d = 0.775$; hyperactivity: $M = 3.42$, $SD = 2.5$, $t(52)$-2.65, $p = .011$, $d = 0.729$; Table 2). Although individuals with ADHD on average met the clinical threshold for current inattention on the BAARS-IV ($M = 5.57$, $SD = 2.71$), while those without ADHD did not ($M = 4.87$, $SD = 2.75$), this was not a statistically significant difference ($t(52) = 0.92$, $p = .963$, $d = 0.254$). On the CAARS, individuals with ADHD displayed significantly more problems with self-concept ($M = 63.08$, $SD = 11.54$) than those without ADHD ($M = 49.07$, $SD = 19.06$; $t(26) = 2.31$, $p = .029$, $d = 0.874$), and scored significantly higher on the ADHD Index (ADHD: $M = 68.08$, $SD = 9.60$; no ADHD: $M = 57.20$, $SD = 15.20$; $t(26) = 2.23, p = .034, d = 0.846$). Both those with ($M = 77.4$, $SD = 13.29$) and without ADHD ($M = 65.67$, $SD = 18.22$) on average met DSM-IV ADHD criteria and DSM-IV inattentive symptoms (ADHD: $M = 79.31$, $SD = 10.39$; non-ADHD: $M = 68.87$, $SD = 23.51$) on the CAARS. Only individuals with ADHD, on average, met DSM-IV criteria for hyperactive symptoms ($M = 65.08$, $SD = 10.58$) and endorsed clinically significant inattention and memory problems ($M = 71.85$, $SD = 11.58$). Although this outcome was not statistically significantly different from those without ADHD ($M = 55.13$, $SD = 16.37$; $M = 61.80$, $SD = 16.70$; $t(26) = 1.87$, $p = .072$, $d = 0.710$, $t(26) = 1.82$, $p = .080$, $d = 690$, respectively),

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Table 1 Sample demographics by cohort type.

Demographic	ADHD	No ADHD	Statistic	Effect-size	*p*
Age (years), *M*(*SD*), t, *d*	31.97(7.87)	33.97(9.58)	0.922	0.227	.555
Female sex, *n*(%), χ^2, phi	20(66.67)	29(78.38)			
Gender Identity, Cramer's V			-	0.194	.279
Female, *n*(%)	18(60.00)	28(75.68)			
Male, *n*(%)	9(30.00)	8(21.622)			
Nonbinary, *n*(%)	3(10.00)	1(2.70)			
Race, Cramer's V			-	0.306	.258
Asian, *n*(%)	1(3.33)	3(8.12)			
Black/African American, *n*(%)	1(3.33)	4(10.81)			
Hispanic/Latinx, *n*(%)	2(6.67)	8(21.62)			
Middle Eastern or North African, *n*(%)	1(3.33)	1(2.70)			
White, *n*(%)	23(76.67)	20(54.05)			
More than one race, *n*(%)	2(6.67)	1(2.70)			
Handedness, Cramer's V			-	0.142	.674
Left-handed, *n*(%)	3(10.00)	3(8.12)			
Right-handed, *n*(%)	26(86.67)	33(91.89)			
Ambidextrous, *n*(%)	1(3.33)	0(0)			
Years of education, *M*(*SD*), t, *d*	15.60(2.06)	15.95(1.86)	0.722	0.177	.473
Premorbid IQ estimate, *M*(*SD*), t, *d*	110.10(9.50)	110.81(8.81)	0.313	0.077	.247

$^{***}p < .001$; $^{**}p < .01$; $^{*}p < .05$; Note: independent samples t-tests were used to determine statistically significant differences between groups for continuous variables, and Chi Square test of independence and Fisher's Exact Test were used for categorical variables.

the effect sizes were moderate. Effect sizes for all subscales were > 0.6, apart from DSM-IV inattentive symptoms.

Five individuals' BDI-II scores were removed from the analysis due to SVT failure. The remaining participants with and without ADHD met clinically significant thresholds on the BDI-II and GAD-7 (Table 2). However, individuals with ADHD endorsed significantly more symptoms of anxiety ($M = 14.60$, $SD = 4.55$), than those without ADHD ($M = 9.00$, $SD = 4.92$; $t(28) = 3.01$, $p = .005$, $d = 1.17$). On the PAI, both those with and without ADHD met the clinical threshold on the following clinical scales: Anxiety, Depression, Schizophrenia, and Borderline. Only those without ADHD met the clinical threshold for anxiety-related disorders, but this difference was not statistically significant. All effect sizes were < 0.6 and no differences in scores were statistically significant.

Aim 3. Sensitivity and specificity analyses

As there were no statistically significant differences in demographic characteristics and covariates, no variables other than each test being evaluated were included in the logistic regression models. Logistic regressions revealed that no cognitive measures were predictive of ADHD diagnosis (Supplementary Table 4). The Generalized Anxiety Disorder-7 was the strongest predictive scale of ADHD diagnosis (AUC = 0.800; $\chi^2(28) = 8.216$, $p = .004$, Nagelkerke $R^2 = 0.333$; Fig. 3) and correctly predicted 76% of cases. Adults who endorsed higher levels of anxiety were 28% more likely to be diagnosed with ADHD (OR = 1.28, 95%CI [1.05, 1.57]). ADHD scales were less predictive of diagnosis. The most predictive clinical scale of the ADHD surveys was CAARS Self-Concept, (AUC = 0.750; $\chi^2(27) = 5.436$, $p = .020$, Nagelkerke $R^2 = 0.236$; Fig. 3), which correctly predicted 75% of cases. Nonetheless, individuals who scored higher on this clinical scale were only 7% more likely to be diagnosed with ADHD (OR = 1.07, 95%CI [1.00–1.13]). Additional clinical scales that predicted ADHD diagnosis, in order of highest sensitivity and specificity to lowest, include the following: Childhood Hyperactivity symptom counts on the BAARS-IV (AUC = 0.712; $\chi^2(53) = 6.682$, $p = .010$, Nagelkerke $R^2 = 0.156$), which correctly predicted 65% of cases and in which individuals who endorsed more childhood hyperactivity symptoms were 36% more likely to be diagnosed with ADHD (OR = 1.36, 95%CI [1.06–1.75]); the ADHD Index on the CAARS (AUC = 0.708, $\chi^2(27) = 5.169$, $p = .023$, Nagelkerke $R^2 = 0.225$), which predicted 61% of cases and in which individuals with a greater ADHD Index were 8% more likely to be diagnosed with ADHD, and Childhood Inattention symptoms counts on the BAARS-IV (AUC = 0.706; $\chi^2(53) = 7.527$, $p = .006$, Nagelkerke $R^2 = 0.175$), which predicted 69% of cases and in which individuals who endorsed more childhood inattentive symptoms were 33% more likely to be diagnosed with ADHD (OR = 1.33, 95%CI [1.07–1.66]; Fig. 3). DSM-IV Hyperactive Symptoms on the CAARS had an AUC of 0.700, but the model overall was not statistically significant ($\chi^2(27) = 3.615$, $p = .057$, Nagelkerke $R^2 = 0.162$). No PAI clinical scales were predictive of ADHD diagnosis (Supplementary Table 4).

Discussion

This study examined adults referred for ADHD evaluation within an outpatient psychiatric neuropsychological setting, where there was substantial overlap between individuals diagnosed with ADHD and other psychiatric conditions. ADHD

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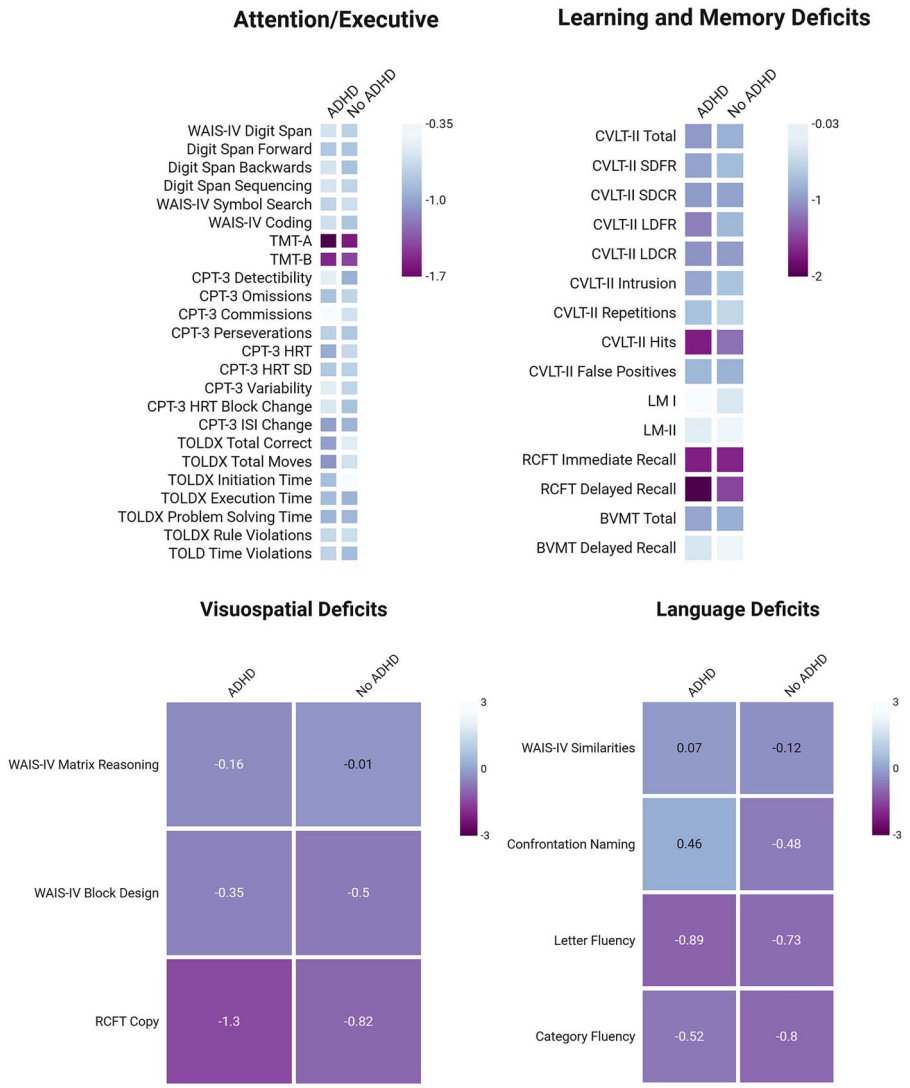


Figure 2 These heat maps demonstrate the average deviation z scores on each neuropsychological test evaluated by group: ADHD and non-ADHD. Z-scores of -1.5 and under represent testing deficits. Darker colors indicate more negative average z scores. Created in BioRender. Pleshkevich, M. (2026) https://BioRender.com/eq8mddd.

referrals represented a large proportion of clinic presentations, and psychiatric comorbidity was highly prevalent across the sample, regardless of diagnostic outcome. Cognitive testing revealed broadly similar performance patterns between groups, with a few meaningful differences and shared weaknesses in aspects of attention and visuospatial incidental memory. In contrast, self-reported symptom history and emotional functioning, particularly anxiety and childhood ADHD, were more informative in distinguishing diagnostic outcomes than objective cognitive measures. Collectively, these findings suggest that adults presenting for ADHD evaluation represent a clinically complex population in which emotional distress and overlapping psychiatric symptoms play a central role, highlighting the importance of comprehensive, multimethod assessment approaches that emphasize developmental history and psychological context.

Aim 1: ADHD evaluation clinic summary

A large proportion of the referrals to the clinic were for ADHD evaluations, suggesting that concerns about attention and executive functioning are a major reason adults seek neuropsychological assessment. For many years, ADHD was viewed as a childhood condition. It is now recognized that symptoms continue into adulthood, which means that many adults who struggled earlier in life were never evaluated. National data revealed that a large portion of adults with ADHD were diagnosed after age 18, suggesting that many cases were missed rather than newly developed (Staley et al., 2024). Modern life, including workplaces and daily routines, now demands high levels of sustained attention, organization, multitasking, and self-management, which can expose weaknesses that previously went unnoticed. Even people without ADHD may feel stressed, which leads some to wonder whether ADHD is the explanation. Additionally, symptoms like poor concentration, forgetfulness, and mental fatigue also are common in anxiety and mood disorders, so many adults experiencing emotional distress may wonder if ADHD explains their difficulties. Emotional distress can make attentional problems feel similar to ADHD, prompting evaluations. This level of referrals appears to be consistent with findings from

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Table 2 Differences in ADHD and psychological screeners by group.

Screener	ADHD, *M*(*SD*)	No ADHD, *M*(*SD*)	t	*d*	*p*
BAARS-IV (raw score symptom count; $N = 55$)					
Current Inattention	5.57(2.71)✣	4.87(2.75)	0.922	0.254	.361
Current Hyperactivity	4.52(2.00)	3.52(2.49)	1.593	0.438	.117
Childhood Inattention**	5.87(2.38)✣	3.77(2.92)	2.816	0.775	.007
Childhood Hyperactivity*	5.13(2.12)✣	3.42(2.5)	2.650	0.729	.011
CAARS (T scores; $N = 29$)					
Inattention/Memory Problems	71.85(11.58)✣	61.80(16.70)	1.830	0.690	.080
Hyperactivity	61.62(9.16)	54.60(12.60)	1.661	0.630	.109
Impulsivity	61.62(13.85)	51.27(14.43)	1.927	0.730	.065
Self-concept*	63.08(11.54)	49.07(19.06)	2.306	0.874	.029
DSM-IV Inattentive Symptoms	79.31(10.39)✣	68.87(23.51)✣	1.478	0.560	.151
DSM-IV Hyperactive Symptoms	65.08(10.58)✣	55.13(16.37)	1.874	0.710	.072
DSM-IV ADHD Total ($N = 26$)	77.40(13.29)✣	65.67(18.22)✣	1.745	0.713	.094
ADHD Index*	68.08(9.60)✣	57.20(15.10)	2.233	0.846	.034
BDI-II ($N = 34$)	12.90(6.64)✣	15.96(8.04)✣	1.059	0.399	.298
GAD-7 ($N = 31$)**	14.60(4.55)✣	9.00(4.92)✣	3.009	1.17	.005
PAI (T scores; $N = 51$)					
Inconsistency	51.39(1.98)	51.11(8.10)	0.115	0.032	.909
Infrequency	52.35(9.05)	53.64(7.99)	0.543	0.153	.590
Negative Impression Management	55.22(12.05)	55.14(14.12)	0.020	0.006	.984
Positive Impression Management	37.39(8.34)	38.61(11.38)	0.427	0.120	.671
Somatic Complaints	53.82(8.62)	55.39(10.53)	0.568	0.162	.573
Anxiety	63.73(14.04)✣	66.86(14.30)✣	0.774	0.221	.443
Anxiety-Related Disorders	59.36(15.08)	63.46(14.80)✣	0.964	0.275	.340
Depression	65.23(13.65)✣	63.71(13.95)✣	0.384	0.109	.702
Mania	55.95(10.99)	54.57(10.74)	0.447	0.127	.657
Paranoia	52.14(11.71)	53.54(12.37)	0.407	0.116	.686
Schizophrenia	60.14(12.37)✣	61.25(11.38)✣	0.331	0.094	.742
Borderline	61.09(8.94)✣	61.32(12.22)✣	0.070	0.020	.944
Antisocial Features	54.11(10.39)	50.11(10.62)	1.436	0.409	.158
Alcohol Problems	53.00(11.51)	49.86(8.98)	1.085	0.309	.283
Drug Problems	52.41(10.71)	49.21(6.72)	1.290	0.367	.203
Aggression	48.55(12.29)	47.00(7.30)	0.553	0.158	.583
Suicide	55.77(13.75)	51.46(12.16)	1.174	0.335	.246
Stress	51.14(8.49)	52.50(12.50)	0.438	0.125	.663
Nonsupport	55.82(14.48)	54.61(14.20)	0.297	0.085	.768
Treatment Rejection	44.64(8.40)	43.57(9.32)	0.332	0.095	.741
Dominance	47.00(9.89)	43.57(9.32)	1.258	0.358	.215
Warmth	47.64(12.82)	47.93(9.63)	0.092	0.026	.926

***$p < .001$; **$p < .01$; *$p < .05$; ✣ Deviation is a clinically meaningful deficit at $z \leq -1.5$ or raw score meets a clinical cut-off. Note: independent samples t-tests were used to determine statistically significant differences between groups.

other outpatient psychiatric settings, where ADHD concerns are common (Adamis et al., 2022; Almeida Montes et al., 2007) and are often under-recognized as part of broader mental health presentations.

Psychiatric comorbidity was highly prevalent among individuals diagnosed with ADHD. This aligns closely with existing research showing that most adults with ADHD also experience additional psychiatric conditions, especially anxiety, mood, or substance use concerns. Some studies found that psychiatric comorbidity is more a given than the exception in adults diagnosed with ADHD, specifically in specialty mental health settings (Choi et al., 2022; Piñeiro-Dieguez et al., 2016). As previously stated, individuals seeking an ADHD evaluation in psychiatry clinics may already have complex emotional sequelae. An important observation found was that people with and without ADHD showed similar rates of mood and anxiety disorders. This mirrors broader mental health clinic populations, where anxiety and

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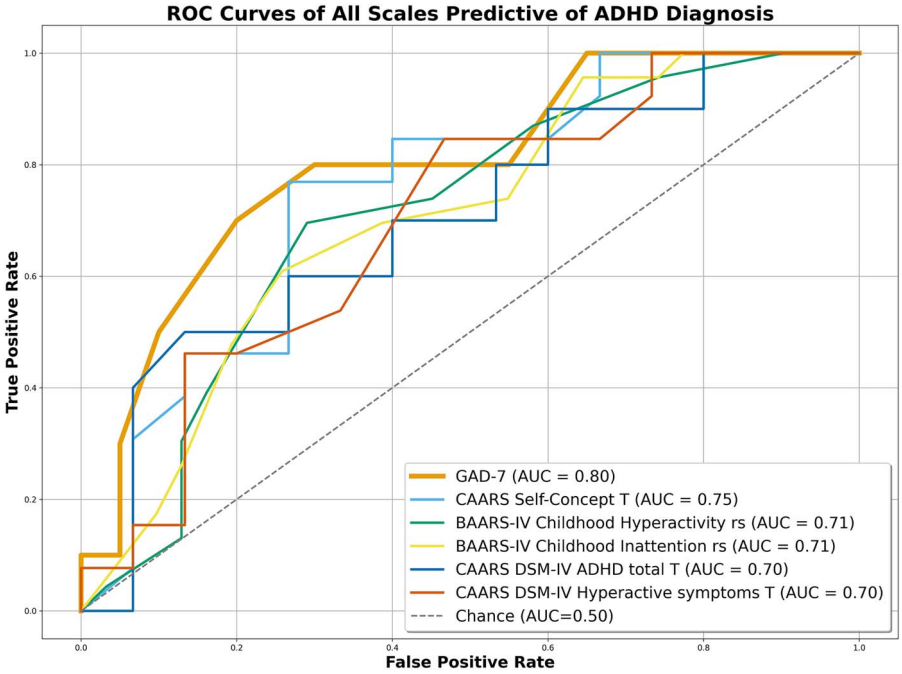


Figure 3 Each line represents a distinct ROC curve analysis rather than a model that includes all variables. Individual ROC curves are presented together in the above figure for comparative clarity. Lines with greater areas under their curve represent measures that are more sensitive and specific in identifying an ADHD diagnosis. Measures are listed in the legend in order of most sensitive and specific to least.

depression are frequent regardless of diagnosis. It suggests that adults referred specifically for ADHD assessment may represent a group with generally high emotional distress, regardless of whether ADHD was diagnosed.

Finally, a concrete ADHD presentation (i.e., inattentive, hyperactive, or combined) was not identified in almost a third of individuals diagnosed with ADHD in the present sample. This could reflect difficulty with retrospective report of childhood ADHD symptoms, which are required for a DSM-5-TR diagnosis, and difficulty parsing out the relative impact of ADHD symptoms versus comorbid psychopathology on daily functioning. For example, a clinician may have been reluctant to specify a specific ADHD presentation if there was significant discordance between childhood and adult symptom report, or if there were unclear patterns of symptoms depending on setting (i.e., inattentive symptoms isolated to work settings but hyperactive symptoms present across work, home, and social settings), particularly in light of psychiatric comorbidity.

Aim 2: Covariates and cognitive and psychological profiles

The ADHD and non-ADHD groups were well matched on many background variables, reducing the likelihood that differences in the clinical outcomes were driven by demographic factors or evaluator bias. Both providers also appeared consistent in diagnostic decision-making, supporting the reliability of clinical evaluations across clinicians. Notably, both groups showed high rates of substance use histories and current marijuana use, which reflects real-world psychiatric populations but may complicate interpretation of cognitive performance. Average abstinence from marijuana was 10 days prior to testing; however, long-term effects on cognition cannot be fully ruled out. Half of the participants identified English as their primary language, which introduces potential cultural and linguistic influences on testing and symptom reporting. Childhood trauma and/or neglect were common in both groups. This is clinically meaningful because trauma history can produce attentional problems, emotional dysregulation, and executive difficulties that overlap with ADHD symptoms, making diagnostic differentiation more challenging. Literature suggests that trauma exposure may increase the likelihood of anxiety and functional impairment in adults presenting with ADHD symptoms (Peleikis et al., 2022).

The most striking finding was that cognitive testing did not clearly differentiate adults with ADHD from those without ADHD who had psychiatric conditions. This notably includes the CPT-3, which is one of the most frequently recommended tests for ADHD (Fuermaier et al., 2019), although an increasing number of studies have been published demonstrating its weak predictive utility (Callan et al., 2024). Both those with and without ADHD showed similar cognitive patterns, and both demonstrated mild difficulties on tasks involving visual scanning speed, organization, and incidental memory. Additionally, both groups reported similar psychological distress on the PAI, with clinically elevated scales relating to thought dysfunction, impulsivity, risk-taking, anxiety, and depression. This merely highlights that disorders such as anxiety, depression, or chronic stress may produce cognitive inefficiencies similar to those of ADHD. In other words, cognitive challenges seen during testing may reflect general psychiatric burden rather than ADHD-specific deficits. This aligns with growing literature suggesting that neuropsychological testing alone

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has limited specificity for adult ADHD in complex clinical samples (Guo et al., 2021; Pettersson et al., 2018).

The effect sizes and cognitive findings in the present study are lower than those reported in prior meta-analyses and systematic reviews on adult-ADHD neurocognitive profiles (Hervey et al., 2004; Onandia-Hinchado et al., 2021). Notably, however, these studies compared those with ADHD to normal healthy controls, rather than psychiatrically matched controls, as the present study did. Existing studies comparing cognitive profiles of those with ADHD to psychiatric controls differ from the present study in that their ADHD groups included those with ADHD alone in addition to individuals with ADHD who meet diagnostic criteria for another DSM psychiatric diagnosis (Rosso et al., 2023). Thus, the ADHD groups in these studies contain less psychiatric burden than the present sample and cannot necessarily be directly compared to the current studies. Nonetheless, Rosso et al.'s (2023) narrative systematic review confirms that the results from these studies are largely aligned with those of the present study, in that, apart from continuous performance tests, they have provided minimal evidence for differences in neuropsychological testing in ADHD when psychiatric controls serve as the comparison group. For example, the effect sizes comparing ADHD to psychiatric controls reported in Guo et al.'s (2021) study ranged from 0.004 to 0.20, and Pettersson et al.'s (2018) effect sizes ranged from 0.001 to 0.104 when controlling for age and IQ, compared to the present study's effect sizes ranging from 0.009 to 0.445. More research is needed to compare individuals with ADHD-psychiatric-comorbidity to psychiatrically matched controls to better understand whether cognitive deficits in ADHD are unique to the condition or instead reflect psychiatric comorbidity.

Although cognitive scales were not different among those with and without ADHD, self-report ADHD scales provided clearer differentiation. Individuals diagnosed with ADHD described more childhood attention and hyperactivity problems and greater concerns about self-concept, consistent with clinical models proposing that long-standing attentional difficulties can negatively influence self-esteem over time. Importantly, current inattentive symptoms were elevated across both ADHD and non-ADHD groups. This finding has strong clinical implications, such as that adults with anxiety, depression, or other psychiatric difficulties may report significant attention problems even in the absence of ADHD. This reinforces the need to interpret symptom questionnaires within a developmental and emotional context rather than in isolation. The pattern of stronger self-concept concerns in the ADHD group may reflect the cumulative impact of lifelong functional struggles, repeated experiences of underachievement, or chronic frustration associated with executive functioning difficulties.

Aim 3: Sensitivity and specificity

Anxiety symptoms emerged as the strongest indicator associated with receiving an ADHD diagnosis, even more than cognitive measures or many ADHD-specific questionnaires. This finding highlights the close relationship between anxiety and adult ADHD presentation. ADHD and anxiety frequently co-occur in adulthood, with comorbidity rates indicating that anxiety symptoms are a clinically meaningful feature of ADHD presentations in adulthood (Kessler et al., 2006; Sobanski, 2006). Several interpretations are possible. Anxiety may develop secondary to chronic attentional and executive functioning difficulties, reflecting years of coping challenges and functional concerns. Moreover, anxiety may be closely linked to the subjective experience of attentional difficulties and executive dysfunction, increasing symptom endorsement during the evaluation. Alternatively, anxiety itself can intensify perceptions of inattention and cognitive inefficiency, leading individuals to seek an ADHD assessment. The overlap likely reflects bidirectional influences rather than a simple causal relationship. Emotional symptoms, specifically anxiety, play a role in adult ADHD presentations clinically and may strongly influence both referral and diagnostic outcomes.

Notably, an additional interpretation is that anxiety could reflect broader psychological distress rather than ADHD-specific pathology. Generalized distress may increase endorsement of symptoms across multiple domains and influence referral patterns. There is consistent evidence that general symptom measures can be predictive of a diagnosis as well as disorder-specific instruments (Arbisi et al., 2012). Additionally, dimensional models of psychopathology emphasize a general "p" factor underlying internalizing and eternalizing psychiatric symptoms (Caspi et al., 2014). Together, these findings suggest that anxiety may play a role in adult ADHD assessment, potentially reflecting both comorbidity and nonspecific distress that influence diagnostic outcomes, supporting prior findings suggesting that psychological symptom questionnaires may lack diagnostic specificity (Arbisi et al., 2012).

Although individual neuropsychological tests may not accurately identify ADHD among psychiatric samples, increasing evidence shows that the apparent heterogeneity in adult ADHD assessment outcomes often observed may reflect underlying distinct neurocognitive phenotypes (Cerny et al., 2025; Leib et al., 2021; McGuigan et al., 2025). These neurocognitive phenotypes in turn may be associated with comorbid psychopathology, although the evidence is mixed. For example, Leib et al.'s (2021) latent profile analysis (LPA) in a cohort of individuals with ADHD completing the WAIS-IV identified four classes of neuropsychological performance: low general cognitive performance, average performance, high working memory, and high processing speed. The high working memory group self-reported significantly fewer symptoms of depression and anxiety than the other groups. However, Cerny et al.'s (2025) LPA of individuals diagnosed with ADHD found similar rates of elevated comorbid psychopathology across neurocognitive classes. Individuals in the "inattentive" class, marked by worse performance on WAIS-IV Coding, TMT-A, and certain subscales on the CPT-3, self-reported significantly more symptoms of depression than individuals in the "Low Learning/Memory" class and more symptoms of anxiety than the "High Average" class, although these effect sizes were small. This finding mirrors the present results, which found that individuals with ADHD and psychiatric conditions performed below expectation on TMT-A—although the results in the present study were not unique to ADHD and could reflect psychopathology instead.

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McGuigan et al.'s (2025) study is perhaps most comparable to the present study, as their LPA included individuals with and without ADHD who presented for neurocognitive evaluations in a clinical setting. Their identified "Neuropsychiatrically Distressed" class most closely resembles our study's cohort, which included individuals both with and without ADHD with elevated inattentive, hyperactive, and mood scores and overall worse cognitive performance relative to other classes. Interestingly, this class was less likely to be diagnosed with ADHD than other classes. The present study addresses the clinical question of how individuals within this class can be accurately classified as having or not having ADHD based on neuropsychological testing.

It is possible that the present sample is comprised of individuals within one of the aforementioned neurocognitive phenotypes, as perhaps driven by psychopathology. Alternatively, the present sample may contain a number of distinct neurocognitive phenotypes that cannot be segmented due to limits on statistical power, which could account for the lack of observed differences between individuals with and without ADHD in the present clinical sample. Future studies should expand upon existing LPA studies to investigate whether observed profiles in ADHD that are associated with comorbid psychopathology are also present in non-ADHD clinical samples, and whether the presence of ADHD differentiates cognitive functioning among these groups specifically.

Limitations

The present study has several limitations that should be considered when interpreting the findings. First, the final sample size following eligibility confirmation was relatively small ($n = 67$), which may limit the generalizability of results. Because all participants were evaluated at an academic medical center rather than in private practice or community settings outside the Boston area, the sample may not fully represent the broader population of adults assessed for ADHD. Second, there is the potential for misclassification bias, such that some participants may have been inaccurately categorized based on self-report measures or subtle clinician bias during the diagnostic interview. Third, unmeasured confounding variables, including behavioral observations, psychosocial factors, and aspects of clinical history not captured in the dataset, may have influenced diagnostic outcomes. Fourth, parental psychiatric/psychological conditions and outcomes among adults aged 65 years and older were not examined, as these factors were beyond the scope of the study. Future research should investigate their potential role in ADHD-related outcomes across the lifespan. Finally, there may have been bias in the results as driven by the types of assessments that each clinician chose for each individual, and which results, such as failed SVTs, each clinician chose not to include in the report, rendering these raw data to be unavailable.

The use of real-world archival data was a strength of this study, as it adds to a body of literature of present-day neuropsychological practices in adult ADHD evaluations. Nonetheless, the nature of this study lends itself to constraints on generality. For one, given the study's archival nature, we were unable to test less-frequently used materials in ADHD diagnosis that could have performed better in differentiating ADHD from psychiatric cognitive deficits. Given this study's findings, future studies should augment clinical neuropsychological assessments with different diagnostic scales and validate the present study's findings in an intentionally recruited more socio-demographically diverse sample.

The participants on whom the present assumptions are based could be a reflection of socio-demographic factors in adult ADHD referrals rather than the actual nature of ADHD in adulthood across a diverse set of individuals. Included participants were predominantly White women with at least some college level education and with an average estimated IQ (based on word-reading) falling in the 75th percentile. Given that all individuals presenting to our clinic were screened, these demographics reflect the demographics of patients most commonly referred for ADHD who are also diagnosed with comorbid psychiatric disorders. The demographics in the present study align with those presented in systematic reviews on adult ADHD presentations (Adamis et al., 2022; Faheem et al., 2022). For example, although ADHD has been demonstrated to be more prevalent among males (Faheem et al., 2022), other studies of adult ADHD outpatient clinic referrals have similarly found an increased rate of female referrals (Adamis et al., 2022). This may be because girls are underdiagnosed in childhood (Tapia et al., 2024), and symptoms may become more impactful in adulthood due to increased psychological comorbidity (Faheem et al., 2022). Similarly, the high level of education and IQ could account for missed diagnoses in childhood.

Alternatively, relative privilege and bias within our hospital network may be reflected in our sample. While the clinic from which data were collected accepts both public and private insurance, it is unlikely that individuals without insurance coverage and substantial financial means were included in the study. Additionally, the clinic does not accept self-referrals, meaning that physician bias may have played a role in who was referred for an assessment in the first place. Other epidemiological studies have found that racially and ethnically minoritized adults are less likely to have documented medical visits related to ADHD (Adams et al., 2024), perhaps reflecting either cultural differences in symptom perception and engagement with the medical system to address symptoms, or physician bias in taking patient complaints seriously. In fact, studies have found that primary care providers refer Black patients disproportionally less to specialists than White counterparts (Landon et al., 2021). Additionally, racial bias may account for more Black individuals being diagnosed with ADHD in childhood, while White individuals have been found to be disproportionally diagnosed as adults (Shalaby et al., 2024). Finally, ADHD is underdiagnosed in girls, leading to increases in adult diagnoses in women coupled with psychological sequelae (Attoe & Climie, 2023).

Given the above, reflexivity is important when interpreting our study's findings. Rather than using our results to reinforce ADHD diagnostic stereotypes, it is important to evaluate them within the specific socio-demographic context in which the data were collected. It may not be the case that educated White women are more likely to have concerns about ADHD in adulthood over and above individuals with other demographics, but rather that on average, this demographic group is more likely to be referred to and have the means

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to undergo a time-consuming comprehensive neuropsychological evaluation. Thus, the study findings may only apply to a population with presumed relative racial and economic privilege.

Clinical implications and future directions

The study findings have significant clinical implications for assessment of ADHD in outpatient psychiatric settings. The overlap of cognitive performance between individuals diagnosed with ADHD and those with other psychiatric conditions suggests that neuropsychological test results alone have limited specificity in complex clinical samples. Shared weaknesses in attention and incidental memory appear to reflect transdiagnostic inefficiencies rather than disorder-specific deficits. In contrast, developmental history, particularly evidence of childhood ADHD symptoms, and current emotional functioning were more informative in distinguishing diagnostic outcomes. This highlights the importance of conceptualizing ADHD as a neurodevelopmental condition that requires evaluation of longitudinal symptom course, cross-situational impairment, and functional impact over time. Collectively, these findings support a comprehensive, multimethod assessment approach that prioritizes clinical interview, developmental history, and psychological context, with cognitive testing serving as an adjunctive tool for characterization and treatment planning rather than a standalone diagnostic determinant. Future research using longitudinal designs is recommended to better understand additional factors that may contribute to ADHD symptoms, including sleep quality and collateral developmental histories. Although the present study did not identify differences in associated covariates among the ADHD and non-ADHD cohort, such approaches may help clarify the directionality between factors that may affect cognition, such as sleep quality and ADHD-related symptoms, as well as the extent to which sleep-related cognitive difficulties and other modifiable factors may improve with appropriate treatment. Along these lines, neurobiological mechanisms may be an emerging area of research looking at sleep quality and ADHD symptomology, specifically longitudinal methodologies in a larger sample. More general future studies may also focus on generating broad treatment recommendations based on shared cognitive deficit patterns in individuals with ADHD and comorbid psychiatric conditions, as well as developing more sensitive tools to distinguish diagnoses.

Supplementary material

Supplementary material is available at *Archives of Clinical Neuropsychology* online.

Conflict of Interest

We have no known conflict of interest to disclose.

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